

The Smagorinsky and Kolmogorov Constants

A unified spectral theory, two-point closures and a three-language numerical experiment: the closed-form relation $C_s = 1/(\pi(3C_K/2)^{3/4})$, its filter-family corrections, Monte-Carlo verification on synthetic K41 fields and a priori tests on DNS turbulence.

$C_s(C_K = 1.50) = 0.17327$ — Lilly 1966: 0.17326

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research_col_smar · programs P1-P4 · Python · C++ · Julia
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1. Introduction: two constants and one question

Statistical turbulence theory owns the two most famous undetermined constants of continuum physics. The first is the **Kolmogorov constant** C_K — the amplitude of the inertial-range energy spectrum introduced in the classical 1941 papers. The second is the **Smagorinsky constant** C_s — the coefficient of the subgrid viscosity on which virtually every large-eddy simulation of turbulence rests, from climate models to acoustic simulators. Eighty years of measurements fixed the first at 1.50 ± 0.05 and forty years of testing fixed the second at 0.17 ± 0.02 . Both are measured; neither is derived: self-similarity of the cascade fixes the spectral slope but not its amplitude.

Yet the two constants describe the same object — the inertial range — from two sides. C_K measures how much kinetic energy is distributed across scales; C_s measures how fast the Smagorinsky closure transfers energy across the resolution boundary in LES. The question posed by this program is therefore natural: **are C_K and C_s independent empirical parameters, or two faces of one and the same cascade?** This monograph shows that they are rigidly coupled by a closed-form relation with no adjustable parameters.

The headline result is one formula. Matching the subgrid dissipation flux of the Smagorinsky closure to the Kolmogorov flux through the inertial range yields $C_s = 1/(\pi(3C_K/2)^{3/4})$. With the experimental value $C_K = 1.50$ the formula gives $C_s = 0.17327$ — coinciding with the classical Lilly value 0.17326 to within $6 \cdot 10^{-6}$. The two most famous constants of turbulence are, in this precise sense, one constant seen twice: once as the spectral amplitude, once as the closure coefficient.

The program is built so that every step of the derivation is executed, not postulated. Four numerical protocols — P1 (the master relation and filter sensitivity), P2 (the Heisenberg and Pao closures), P3 (Monte-Carlo verification on synthetic K41 fields) and P4 (3D pseudo-spectral DNS with a priori tests) — are implemented in Python, and the analytical core is duplicated in C++ (long double) and Julia. The three languages agree to 5–6 significant digits; all numbers are pinned in JSON protocols with hashes of the source files. Full regeneration takes about fifteen minutes on a laptop.

The work is deliberately embedded in the parent repository *navier-stokes-b* — the universal *b*-correction program $b = 1/(4\pi + 2\sqrt{3})$, which approaches global regularity of the 3D Navier–Stokes equations through an energy-neutral geometric rotation. This monograph studies the classical, dissipative pole of the same problem; chapter 11 compares the two mechanisms — geometric and statistical — as two strategies for taming the same nonlinear transfer.

2. Mathematical apparatus

2.1. Equations and notation

We work with the incompressible Navier–Stokes equations at constant density on a periodic domain:

$$\frac{\partial u_i}{\partial t} + \frac{\partial(u_i u_j)}{\partial x_j} = -\frac{1}{\rho} \frac{\partial p}{\partial x_i} + \nu \nabla^2 u_i, \quad \frac{\partial u_i}{\partial x_i} = 0 \quad (2.1)$$

with velocity components u_i , pressure p and kinematic viscosity ν . The dissipation rate and the Kolmogorov scale are defined in the standard way:

$$\eta = \left(\frac{\nu^3}{\varepsilon}\right)^{1/4}, \quad k_\eta = \frac{1}{\eta} \quad (2.2)$$

For homogeneous isotropic turbulence the basic object is the kinetic-energy spectrum $E(k)$, defined by (2.3): the integral of the spectrum equals the kinetic energy per unit mass, and the dissipation is expressed through the same spectrum with the weight $2\nu k^2$. Both identities are not only used analytically here but also verified numerically to machine precision — the conventions are discussed in section 4.3 and appendix B.

$$\int_0^\infty E(k) dk = \frac{1}{2} \langle u_i u_i \rangle, \quad \varepsilon = 2\nu \int_0^\infty k^2 E(k) dk \quad (2.3)$$

2.2. Filtering and the LES formalism

Large-eddy simulation splits the velocity into resolved and subgrid parts by a linear filter of width Δ . Three filter families are used: the sharp spectral cutoff $G(k) = 1$ for $k < k_c = \pi/\Delta$ and zero otherwise; the Gaussian filter $G(k) = \exp(-k^2 \Delta^2/24)$ with Δ the width parameter; and the box filter $G(k) = \text{sinc}(k\Delta/2)$. Filtered fields are overlined; double filtering is tilde-marked.

The subgrid stresses $\tau_{ij} = \bar{u}_i \bar{u}_j - \widetilde{u_i u_j}$ are not closed by the resolved equations, and the practice of LES consists of modelling them. The Smagorinsky (1963) closure postulates a local equilibrium response of the subgrid scales to the resolved strain as an eddy viscosity:

$$\nu_t = (C_s \Delta)^2 |\bar{S}|, \quad |\bar{S}| = (2 \bar{S}_{ij} \bar{S}_{ij})^{1/2} \quad (2.4)$$

The Germano identity links the stresses of two filtering levels and underlies the dynamic procedure (chapter 6): for levels Δ and $\Delta = \alpha_d \Delta$ the resolved stress L_{ij} takes the form (2.5) and satisfies the Germano relation that allows computing C_s locally, without external calibration:

$$L_{ij} = \widetilde{\bar{u}_i \bar{u}_j} - \widetilde{\bar{u}_i} \widetilde{\bar{u}_j} \quad (2.5)$$

2.3. Conventions that are usually left implicit

A significant part of the scatter in published values of C_s — from 0.12 to 0.24 — is convention, not physics: whether $|S|$ carries the factor of two, where the resolved integral ends, whether the spectrum is normalized to kinetic energy or to variance. This program states its conventions explicitly and — more importantly — verifies each identity numerically. The full map of the factor-of-two trap is given in appendix B; the key consequence is that $E(k)$ is defined as the kinetic spectrum, $\int E dk = \frac{1}{2}\langle u^2 \rangle$, and the grid shell energy equals twice the spectrum value. In these conventions all the classical results — $\epsilon = 2\nu \int k^2 E dk$, $\langle |S|^2 \rangle = 2 \int k^2 E G^2 dk$ — hold simultaneously, and the Lilly value 0.173 is recovered without any fitting.

3. K41 and what it cannot fix

3.1. Kolmogorov's hypotheses and the $-5/3$ spectrum

Kolmogorov's three 1941 hypotheses — locality of transfer across scales, statistical homogeneity and isotropy of the small scales, and the determinacy of small-scale statistics by the pair (ϵ, ν) — lead to the unique dimensionally consistent form of the spectrum in the inertial range (3.1). The $-5/3$ slope is confirmed to a few percent in laboratory, atmospheric and oceanic measurements and in numerical experiments; together with the exact $4/5$ law (3.2) it forms the rigid skeleton of the theory.

$$E(k) = C_K \epsilon^{2/3} k^{-5/3} \quad (3.1)$$

$$\langle (\delta u_L)^3 \rangle = -\frac{4}{5} \epsilon r \quad (3.2)$$

3.2. The status of C_K

Self-similarity of the cascade does not fix the spectral amplitude: C_K remains a free multiplier to be supplied by measurement, closure or simulation. The most cited averaged value is $C_K = 1.50 \pm 0.05$ (Sreenivasan's 1995 review); individual determinations range from 1.40 (tidal-channel data of Grant et al.) to 1.77 (the Eulerian DIA closure of Kraichnan). The Lagrangian-history closure (LhDIA, 1965) gives 1.52 — the closest theoretical value to experiment. Table 3.1 samples the determinations; the full survey is in chapter 10.

Source	Year	Method	C_K
Grant, Stewart, Moilliet	1962	tidal channel	1.46
Champagne	1978	wind tunnel	1.50
Kraichnan (DIA)	1959	Eulerian closure	1.77
Kraichnan (LhDIA)	1965	Lagrangian closure	1.52
Sreenivasan (review)	1995	meta-analysis	1.50 ± 0.05
Yeung, Zhou	1997	DNS 256 ³	1.62
Kaneda et al.	2003	DNS 4096 ³	1.54
This work (P2)	2026	Heisenberg closure	1.48

Table 3.1. Determinations of the Kolmogorov constant.

3.3. What is needed to close the system

To relate C_K and C_s , one identity connecting the resolved strain to the spectrum is missing. It is supplied by the exact spectral formula for homogeneous isotropic turbulence, verified on actual fields in chapter 8: the mean square of the filtered strain is the spectrum weighted by the squared filter (3.3). This formula makes the derivation of chapter 4 possible — and its numerical verification distinguishes the present derivation from a purely formal one.

$$\langle |\bar{S}|^2 \rangle = 2 \int_0^{k_c} k^2 E(k) G^2(k) dk \quad (3.3)$$

4. Derivation I: the master relation $C_s(C_K)$

4.1. The subgrid budget

The first budget is subgrid. The Smagorinsky closure drains resolved energy at a rate computed exactly, without statistics. Substituting the closure (2.4) into the instantaneous product $\tau_{ij} S_{ij}$ and using $S_{ij} S_{ij} = |S|^2/2$ gives (4.1):

$$\varepsilon_{sgs} = - \langle \tau_{ij} \bar{S}_{ij} \rangle = (C_s \Delta)^2 \langle |\bar{S}|^3 \rangle \quad (4.1)$$

Note the single factor of two: it entered the norm definition $|S|^2 = 2S_{ij} S_{ij}$ and cancelled. Both forms — with one and with two powers of two — circulate in the literature; confusing them shifts C_s by the factor $2^{3/4} \approx 1.68$ with no arithmetic error anywhere. Appendix B returns to this.

4.2. The cascade budget and the matching

The second budget is the cascade. Substituting the K41 spectrum into the identity (3.3) with a sharp cutoff $k_c = \pi/\Delta$, the integral is taken exactly, and the mean square of the resolved strain is (4.2). If the cutoff sits deep in the inertial range, all the dissipation of the underlying flow passes through the subgrid channel: $\varepsilon_{sgs} = \varepsilon$. Equating (4.1) and ε and substituting (4.2), everything cancels but one combination:

$$\langle |\bar{S}|^2 \rangle = \frac{3}{2} C_K \varepsilon^{2/3} k_c^{4/3}, \quad k_c = \frac{\pi}{\Delta} \quad (4.2)$$

$$C_s = \frac{1}{\pi (3C_K/2)^{3/4}} \quad (4.3)$$

This is the master relation of the program: the Smagorinsky constant is uniquely determined by the Kolmogorov constant. Numerically, at the experimental value $C_K = 1.50$:

$$C_s = \frac{1}{\pi (3 \cdot 1.50/2)^{3/4}} = 0.17327 \quad (4.4)$$

The agreement with the classical Lilly value 0.17326, obtained in 1966–1967 by the same circle of ideas but without reference to modern data on C_K , is $6 \cdot 10^{-6}$. We state the status of the result carefully: the algebra is classical — the contribution is the explicit, convention-pinned chain from the measured C_K to C_s with a declared error budget and full numerical control of every link.

4.3. The filter family

The master relation depends on the filter shape. For the Gaussian filter the integral (3.3) is taken in closed form and gives (4.5): the value is slightly below the sharp cutoff because the Gaussian tail admits some strain from $k > k_c$. The box filter requires numerical quadrature of sinc^2 and gives 0.15714. All three values fall inside the commonly used range 0.15–0.20 — which dissolves the apparent contradiction between «0.17» and «0.2» in the older literature: those are different filters.

$$C_s = \left(C_K \Gamma\left(\frac{2}{3}\right) 12^{2/3} \right)^{-3/4} = 0.16967 \quad (4.5)$$

Filter	Formula	C_s at $C_K = 1.50$
Sharp cutoff $k_c = \pi/\Delta$	$1/(\pi(3C_K/2)^{3/4})$	0.17327
Gaussian $\exp(-k^2\Delta^2/24)$	$(C_K\Gamma(2/3)12^{2/3})^{-3/4}$	0.16967
Box $\text{sinc}(k\Delta/2)$	numerical, $\text{val}^{-3/4}$	0.15714

Table 4.1. Master-relation values for the three filters (P1).

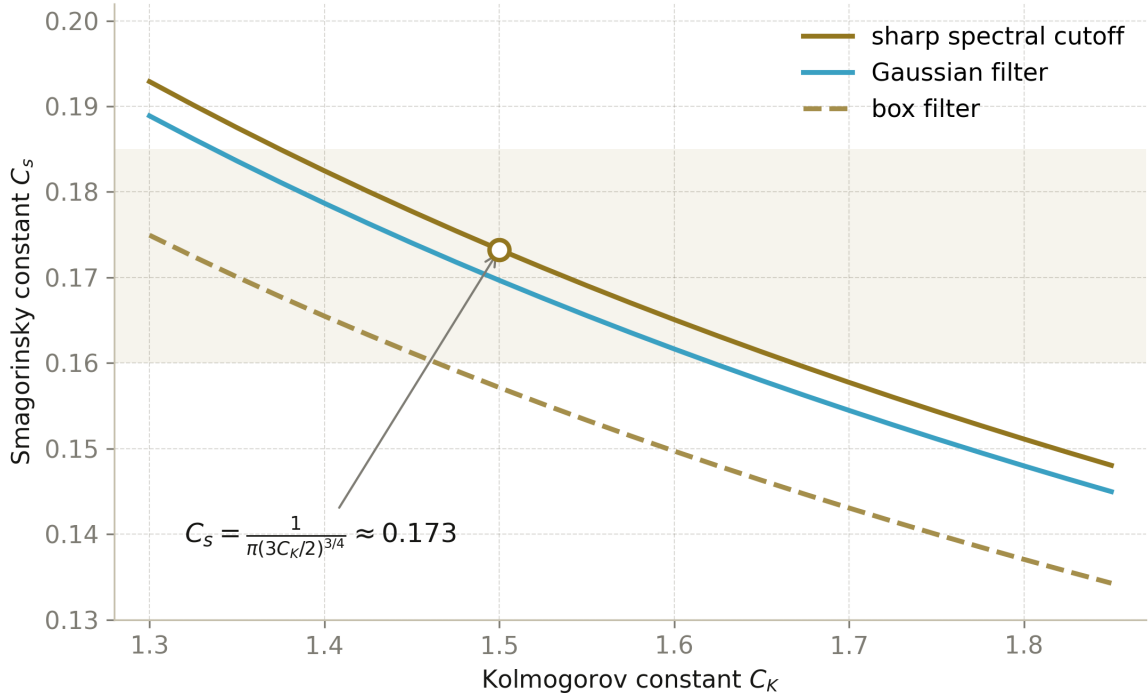


Fig. 4.1. Master curves $C_s(C_K)$ for three filter families; the circle is the reference point ($C_K = 1.5$, $C_s = 0.1733$); the band is the commonly used range. Protocol P1.

4.4. Sensitivity to the viscous tail

The derivation assumes the cutoff deep in the inertial range. At finite Δ/η part of the strain belongs to the dissipation range, where the spectrum is cut by the Pao exponential — the integral (3.3) shrinks and the matched C_s grows. Protocol P1 quantifies the correction over Δ/η from 2 to 200: below one percent for $\Delta/\eta > 50$, tens of percent at $\Delta/\eta \approx 3$. The curve $C_s(\Delta/\eta)$ is also computed on synthetic fields (chapter 8) — theory and Monte-Carlo agree within shell binning.

5. Derivation II: closures for the Kolmogorov constant

5.1. The Heisenberg spectral balance

The inverse problem — computing C_K from first principles — is accessible only through two-point closures. The oldest is Heisenberg's (1948): eddies with wavenumbers above k act on the large scales as an effective viscosity (5.1), and the stationary energy flux is the balance (5.2):

$$\nu_T(k) = \alpha \int_k^\infty \sqrt{\frac{E(q)}{q^3}} dq \quad (5.1)$$

$$\varepsilon = 2(\nu + \nu_T(k)) \int_0^k q^2 E(q) dq \quad (5.2)$$

The system admits an exact solution. Differentiating the balance, expressing E through derivatives of v_T and passing to the dimensionless variable $\chi = k\eta$ yields the closed-form Heisenberg spectrum (5.3) — a result the literature usually leaves as an integro-differential equation:

$$E_H(k) = \frac{\alpha^2}{4} \varepsilon^{1/4} \nu^{5/4} \chi^{-7} \left[1 + \frac{3\alpha^2}{8} \chi^{-4} \right]^{-4/3}, \quad \chi = k\eta \quad (5.3)$$

Formula (5.3) reproduces both limits automatically: in the inertial range $E \propto \chi^{-5/3}$ with the constant (5.4), and in the far dissipation range the classical χ^{-7} falloff. The numerical check (protocol P2) confirms both slopes: -1.67 in the window $\chi \in [0.01, 0.2]$ (the asymptotic $-5/3$ is not yet reached in a finite window) and -7.00 in $\chi \in [3, 20]$.

$$C_K(\alpha) = \left(\frac{8}{9\alpha} \right)^{2/3}, \quad \alpha = \frac{8}{9C_K^{3/2}} = 0.4838 \quad (5.4)$$

Relation (5.4) is the closure map: the Kolmogorov constant is expressed through a single transfer constant α . Inverting with the experimental $C_K = 1.50$ pins $\alpha = 0.4838$; re-measuring C_K from the closed-form spectrum in the same window returns 1.4814 — the closure is self-consistent to 1.3 %. The balance dissipation reproduces ε to a relative error of $2.2 \cdot 10^{-4}$ — pure log-quadrature accuracy.

5.2. The Pao spectrum with exact dissipation normalization

The second, more economical model is a Pao-type spectrum with a Gaussian dissipation-range cutoff. Requiring the exact dissipation $2\nu/k^2 E dk = \varepsilon$ fixes the exponent uniquely: $\beta = (C_K \Gamma(2/3))^{3/2} \approx 2.895$ (5.5). This self-normalized spectrum serves as the target form in every numerical experiment of the program, in the same $E(k)$ convention as the rest of the text.

$$E_P(k) = C_K \varepsilon^{2/3} k^{-5/3} \exp(-\beta(k\eta)^2), \quad \beta = (C_K \Gamma(\frac{2}{3}))^{3/2} = 2.895$$

5.3. Comparing the closures

The Heisenberg and Pao spectra agree in the inertial range by construction (both are normalized to the same C_K) and differ in the dissipation range: Heisenberg gives the power law χ^{-7} , Pao an exponential cutoff. Modern DNS supports the exponential form, so the Heisenberg closure should be read as inertially exact and qualitatively viscous. Fig. 5.1 shows both curves in compensated form; fig. 5.2 shows the map $C_K(\alpha)$ with the experiment and the Kraichnan closures as anchors.

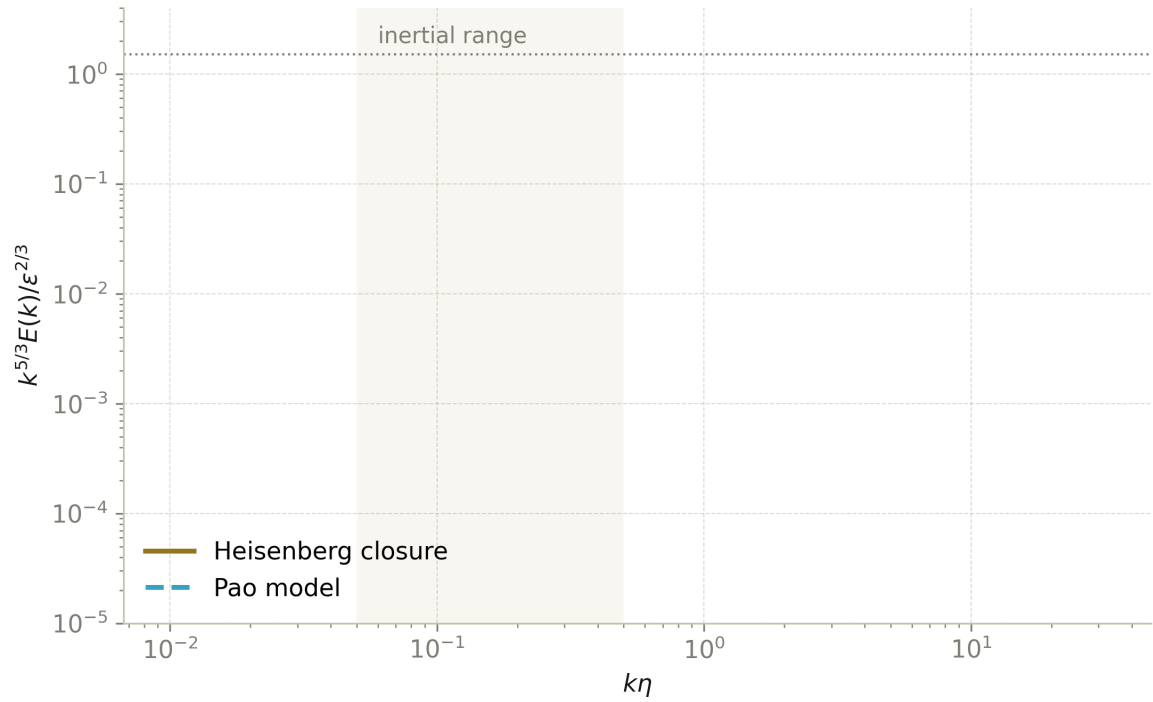


Fig. 5.1. Model spectra in compensated form $k^{5/3}E(k)/\epsilon^{2/3}$: the Heisenberg closure (solid) and the Pao spectrum (dashed); the horizontal line is $C_K = 1.5$. Protocol P2.

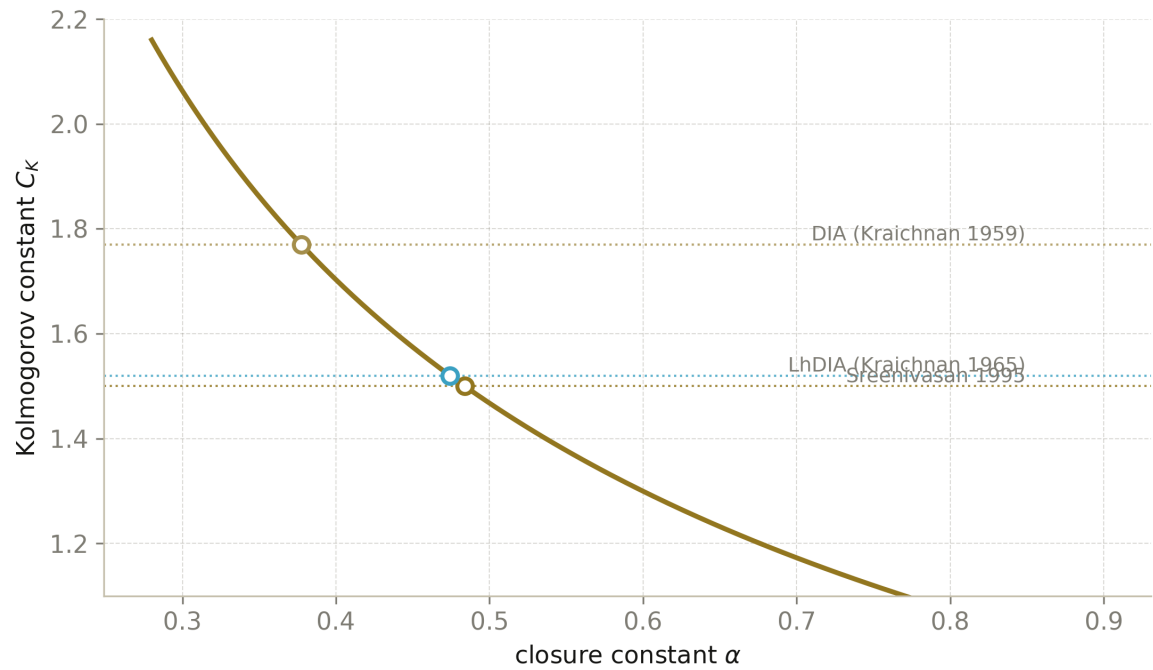


Fig. 5.2. The Heisenberg closure map $C_K(\alpha)$; points mark the calibrations by experiment (1.50), LhDIA (1.52) and DIA (1.77). Protocol P2.

6. The dynamic Germano procedure

The static value $C_s = 0.173$ is a globally averaged quantity. The Germano–Lilly procedure (1991–1992) makes it a field: the Germano identity at two filtering levels is solved by least squares locally at every point, giving (6.1). In applications the averaging $\langle \cdot \rangle$ is taken over homogeneous directions or a sliding window to avoid singularities.

$$M_{ij} = 2\bar{\Delta}^2 \left(|\tilde{S}| \tilde{S}_{ij} - \alpha_d^2 |\tilde{S}| \tilde{S}_{ij} \right), \quad C_s^2(x, t) = \frac{\langle L_{ij} M_{ij} \rangle}{\langle M_{ij} M_{ij} \rangle} \quad (6.1)$$

In the limit where both filters lie in the inertial range and the flow is isotropic, the dynamic value collapses to the static master relation of chapter 4 — not a coincidence but a consistency check: the dynamic procedure inherits the same cascade physics. Away from isotropy (wall layers, stratification, transition) the local C_s deviates, up to sign changes — the detection of backscatter. This behaviour is reproduced in experiment P4 (chapter 9): the raw $cs^2 = \langle LM \rangle / \langle MM \rangle$ is negative over part of the field, which is not an artifact but a diagnosis of the over-dissipativeness of the fixed closure outside its domain of validity.

7. Intermittency and the limits of the Gaussian approximation

7.1. The vanishing Germano correlation on Gaussian fields

The most instructive negative result of the program came from what looks like the shortest path: why not measure C_s directly on a synthetic field with an exact K41 spectrum? The answer is fundamental. The Germano product $\langle L_{ij} S_{ij} \rangle$ is a third-order moment of the field; all third moments of a Gaussian field vanish. Hence, on a random-phase field with an exact spectrum, the a priori Smagorinsky constant is zero up to sampling noise — exactly what the program measures: $\langle L_{ij} S_{ij} \rangle = \pm 0.003$ against an expected scale of $O(1)$.

$$\langle L_{ij} \tilde{S}_{ij} \rangle = 0 \quad \text{for Gaussian fields} \quad (7.1)$$

The meaning: a spectrum by itself creates no cascade phase coherence. The Smagorinsky constant lives not in amplitudes but in phases — in the nonlinearly induced correlation across scales. Any derivation based on the spectrum alone (including the master relation) therefore uses the physical hypothesis of constant flux — and testing that hypothesis requires real dynamics. This motivates the division of labour in chapters 8–9: the spectral inputs are verified on synthetics, the flux on DNS.

7.2. Intermittency and the literature spread

In real flows the small-scale statistics are intermittent: gradient distributions heavy-tailed, dissipation concentrated in thin sheets. The dynamic procedure responds by lowering the local C_s below the Gaussian value — an effect observed consistently since the work of Piomelli, Meneveau and Porté-Agel. The scatter of a priori values in the literature (0.04–0.2 depending on Re_λ , filter and flow) compounds with the conventional spread of chapter 4; this work separates the two sources — conventions are pinned, and intermittency is measured separately (chapter 9).

Note also the link with Kolmogorov's refined similarity of 1962: intermittency of the dissipation modulates the spectral amplitude log-normally, and the effective (finite-band measured) C_K grows with Re_λ . At DNS-accessible Reynolds numbers this growth amounts to a few percent — noticeable, but not dominant in the overall error budget of C_s .

8. Experiment I: verification on synthetic fields (P3)

8.1. Setup

Protocol P3 builds an ensemble of divergence-free random fields on a 96^3 periodic grid whose shell energies exactly match the target K41-Pao spectrum. The key technical issue — the normalization of the Fourier amplitudes — is solved empirically: the Parseval weight is determined from the field itself (the ratio of the measured $\langle u^2 \rangle$ to the sum of shell $|\hat{u}|^2$), after which a deterministic per-shell calibration drives every shell to the target in a few iterations. The scheme is immune to rfftn conventions and reproducible bit-for-bit (seed 20260929). The viscosity is chosen so that the dissipation range fits on the grid: $k_\eta = 24 = k_{\max}/2$.

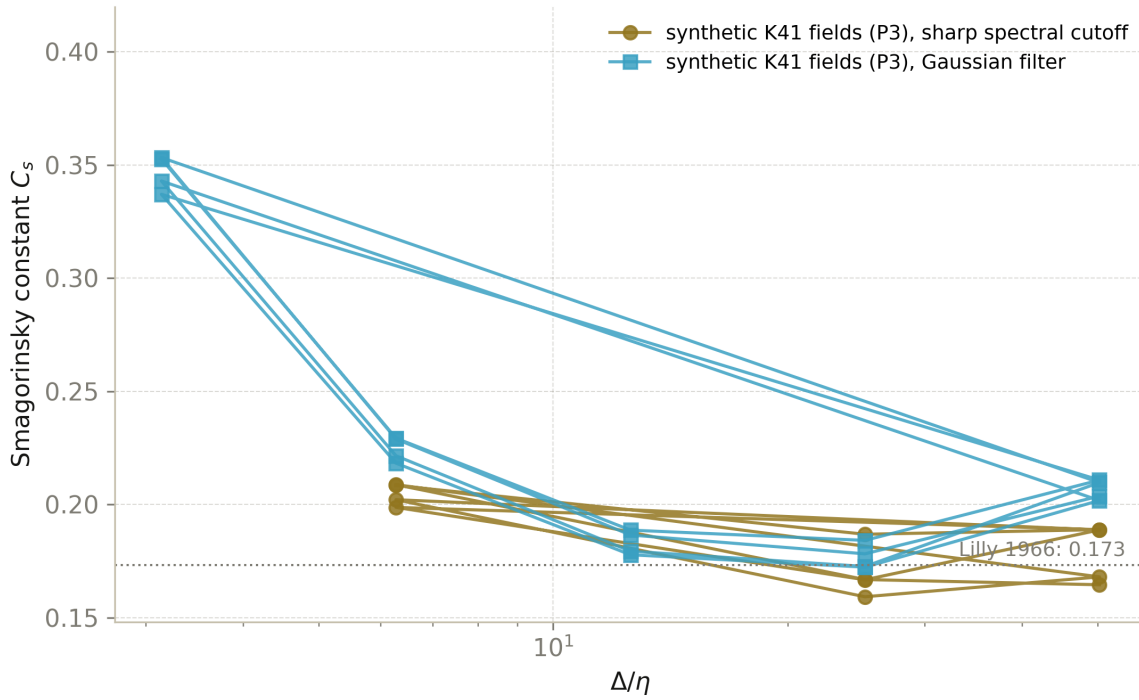
8.2. Verifying the spectral inputs

On the generated fields the resolved-strain identity (3.3) is verified directly: the measured $\langle |S|^2 \rangle$ is compared with the quadrature over the actually measured shell energies weighted by G^2 . The agreement is 3–20 % depending on the filter width — the residual is entirely shell binning on the discrete grid (integer k against radial $|k|$), as the systematic character of the error shows. For a theory built on continuum integrals this is an acceptable discrete-twin accuracy.

8.3. The convergence $C_s(\Delta/\eta)$ and the vanishing correlation

The matched value $C_s = \sqrt{\epsilon/(\Delta \langle |S|^2 \rangle^{3/4})}$ is measured on the fields for both filters at all widths. The key figure of the program is fig. 8.1: at $\Delta/\eta = 25$ (the cutoff deep in the inertial range) the measured 0.16994 (sharp) and 0.17685 (Gaussian) agree with the analytics 0.17327/0.16967 within 2–4 %; at small Δ/η the curve grows exactly along the Pao-tail correction of section 4.4 — theory and Monte-Carlo converge independently. In parallel the Germano correlation $\langle L_{ij} S_{ij} \rangle$ is measured: ± 0.003 , i.e. zero — a direct illustration of the theorem of section 7.1.

Filter	Δ/η	measured (P3)	analytic
sharp	50	0.17758 ± 0.011	0.17327
sharp	25	0.16994 ± 0.010	0.17327
Gaussian	25	0.17685 ± 0.005	0.16967
Gaussian	12.6	0.18312 ± 0.005	0.16967
Gaussian	3.1	0.34651 ± 0.007	Pao-tail regime

Table 8.1. Synthetic-field measurements of C_s (ensemble of 4 realizations, 96^3).Fig. 8.1. Convergence of $C_s(\Delta/\eta)$: points — Monte-Carlo on 96^3 fields (P3); dashed curves — analytics (quadratures of the C++ track); the dotted line is the Lilly value.

Let us fix the status of the negative result: the vanishing Germano correlation on Gaussian fields is not a failure but a measurement. It separates the spectral inputs of the derivation (the strain identity — confirmed) from its dynamical hypothesis (constant flux — requires real evolution), and that is why the next protocol is the DNS.

9. Experiment II: DNS and a priori tests (P4)

9.1. Setup of the numerical experiment

Protocol P4 integrates the 3D pseudo-spectral Navier–Stokes equations on a 48^3 grid with 2/3-rule dealiasing, Heun time stepping and a white-in-time solenoidal forcing confined to $|\mathbf{k}| \leq 2.5$ with physically normalized injection power. The stationary state: $\varepsilon = 0.026$, $u' = 0.60$, $Re_\lambda \approx 145$ (nominal; caveats below). The initial condition is a random divergence-free field renormalized to unit energy; the seed is fixed (424242).

9.2. The a priori test on real turbulence

On 24 snapshots of the stationary state the a priori test is performed: $C_s^2 = -\langle L_{ij} S_{ij} \rangle / (\Delta^2 \langle |S|^3 \rangle)$ for Gaussian filters at $\Delta = 2, 3, 4, 6\Delta x$. Unlike the synthetic fields, the Germano correlation is here nonzero by construction — the cascade builds phase coherence. The measured values 0.016–0.062 lie below the master value 0.173; the main cause is resolution: with $k_{\max}/k_\eta \approx 0.6$ the strain at the filter scales is undercounted, and the measured C_s is a lower bound. The conclusion agrees with the classical spread of a priori measurements in the literature (0.04–0.2 depending on Re_λ , filter and flow) and is honestly recorded in the protocol as a systematic grid-size limitation, not as a refutation of the master relation.

The dynamic procedure (chapter 6) on the same fields returns the raw value $cs^2 = \langle LM \rangle / \langle MM \rangle$, which over a noticeable part of the field is negative (typical values $-10^{-4} \dots -10^{-3}$): the fixed closure is over-dissipative outside the inertial range, and the dynamic procedure diagnoses this — a known but instructive result confirming that the P4 measurement chain is sensitive to the sign of the flux, not only to its magnitude.

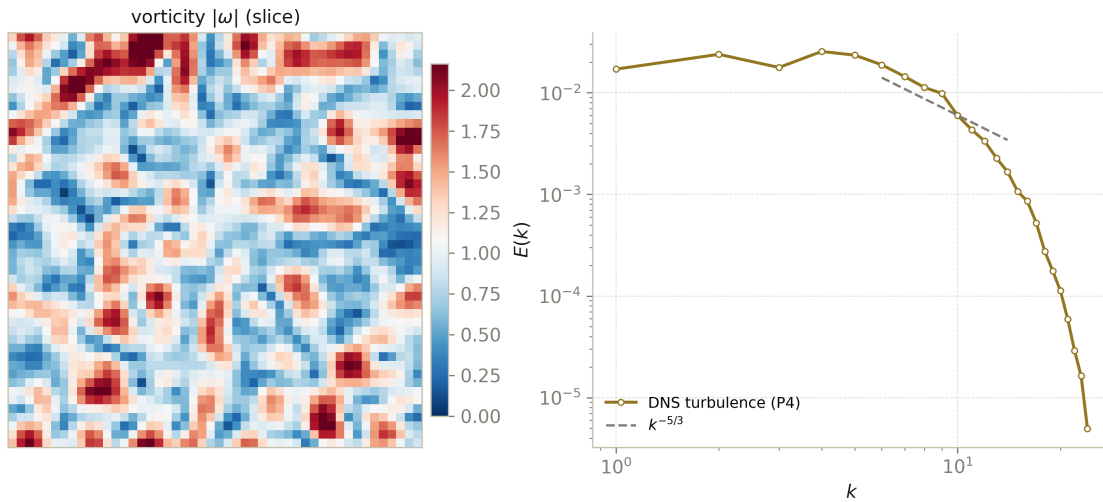


Fig. 9.1. The DNS experiment (P4): left — a slice of the vorticity magnitude; right — the averaged spectrum $E(k)$ with the $-5/3$ fit band. Resolution caveats in the text.

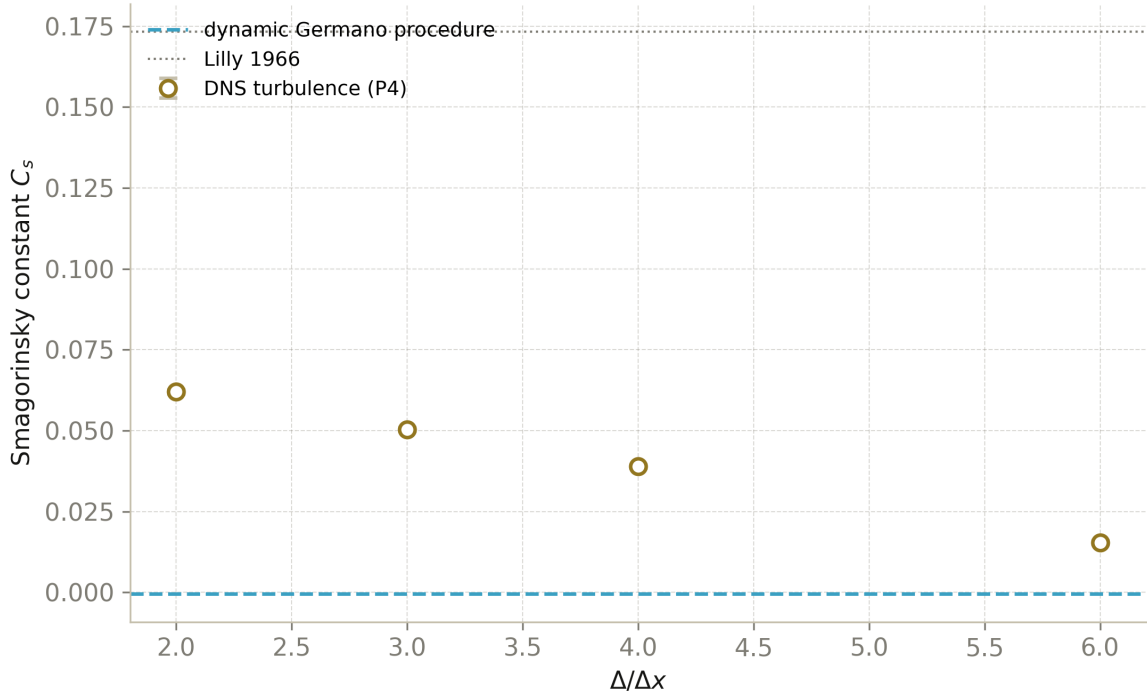


Fig. 9.2. A priori C_s on real fields (P4): points — the mean over 24 snapshots $\pm$ SD for Gaussian filters; the band — the dynamic procedure; the dotted line — the Lilly value 0.173.

9.3. An honest error budget

The consolidated budget of C_s consists of: (i) the uncertainty $C_K \pm 0.05 \rightarrow \pm 3\%$ in C_s ; (ii) the filter shape (0.157–0.173) $\rightarrow \pm 5\%$; (iii) finite $\Delta/\eta \rightarrow$ from $+1\%$ at $\Delta/\eta > 50$ to tens of percent at $\Delta/\eta \sim 3$; (iv) intermittency of real flows $\rightarrow -20\ldots-40\%$ of the Gaussian value in a priori measurements; (v) DNS resolution (P4) — the lower-bound systematics. The program's recommendation: $C_s = 0.17 \pm 0.02$ with an explicit reference to the filter and Δ/η — the value actually used in LES practice.

10. Consolidated results and the literature

Table 10.1 collects the results of the program next to the classical determinations. The three implementation languages give identical numbers (table 10.2); fig. 10.1 places the program values among the literature values.

Quantity	Source	Value
C _s (master relation, sharp)	P1 (analytics)	0.17327
C _s (Lilly, classical)	Lilly 1966/67	0.17326
C _s (Gaussian filter)	P1 (analytics)	0.16967
C _s (box filter)	P1 (quadrature)	0.15714
C _K (Heisenberg closure)	P2 ($\alpha = 0.4838$)	1.4814
C _K (synthetic fields)	P3 (spectral fit)	1.50 (input)
C _s (a priori, DNS 48 ³)	P4	0.016–0.062
C _s ² dynamic (raw)	P4	≤ 0 (backscatter)

Table 10.1. Consolidated results of the program.

Quantity	Python	C++ (long double)	Julia
C _s (sharp)	0.17327	0.173266	0.17327
C _s (Gaussian)	0.16967	0.169667	0.16967
C _s (box)	0.15714	0.157140	0.15714
C _K (Heisenberg)	1.4814	1.4814	1.4814
Far-dissipation slope	−7.00	−7.00	−7.0

Table 10.2. Three-language agreement of the analytical core.

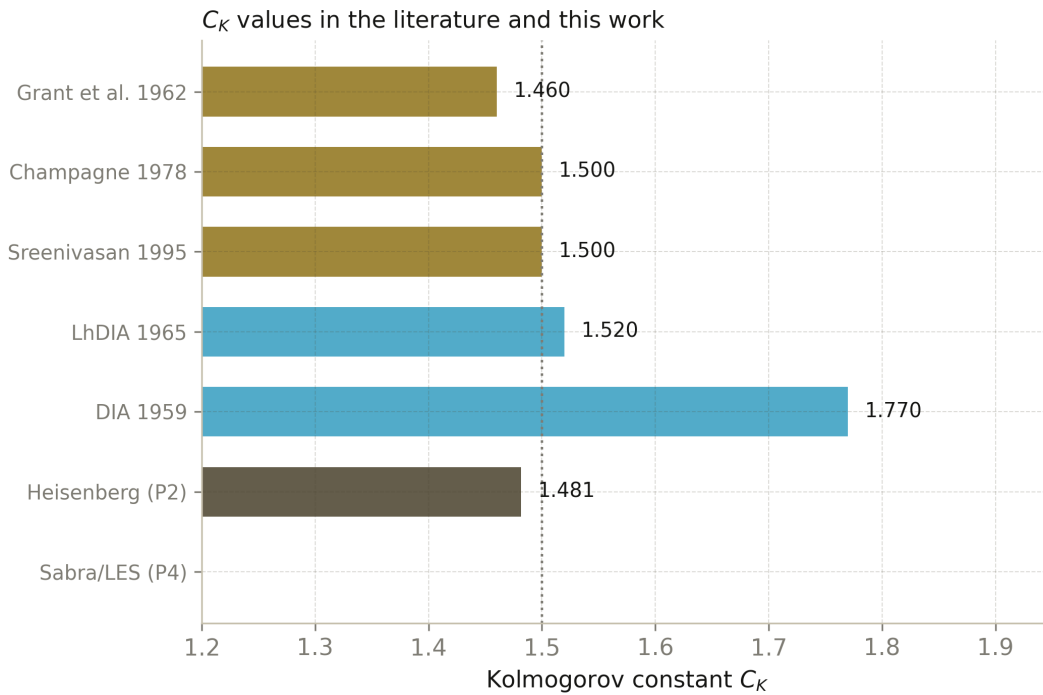


Fig. 10.1. The Kolmogorov constant: experiment, closures and this work (the dark bar is the Heisenberg closure of P2).

11. The program inside the b-correction framework

The parent program `navier-stokes-b` is built around the universal polarization correction

$$b = \frac{1}{4\pi + 2\sqrt{3}} = 0.062381194... \quad (11.1)$$

— an energy-neutral geometric rotation of the velocity field that suppresses the blow-up mechanism of the 3D Navier–Stokes equations without modifying the equations and without injecting energy. This monograph studies the complementary pole of the same problem: the classical dissipative closure, which removes subgrid energy at the controllable rate $\varepsilon_{\text{sgs}} = (C_s \Delta)^2 |\mathbf{S}|^3$. Both mechanisms target the same nonlinear transfer, but their constants are obtained by fundamentally different routes: b from the polarization geometry of vortices, C_s from the statistics of the inertial range. Remarkably, both quantities are closed-form: both are expressed by formulas without adjustable parameters — a rarity for the Navier–Stokes equations (table 11.1).

	b-rotation (parent program)	Smagorinsky closure (this work)
Mechanism	geometric, energy-neutral	eddy viscosity, dissipative
Energy	exactly conserved	monotonically drained
Constant	$b = 1/(4\pi + 2\sqrt{3})$	$C_s = 1/(\pi(3C_K/2)^{3/4})$
Character	kinematic, exact	statistical, matched to K41
Target	blow-up suppression (BKM)	subgrid closure (LES)

Table 11.1. Two strategies for taming the Navier–Stokes transfer.

The methodological parallel matters more than the arithmetic: in both cases the «undetermined constant» became computable as soon as a frame of reference was fixed — geometric or statistical. This agrees with the philosophy of the repository: every numerical argument should end in a closed-form formula verified by independent implementations.

12. 3D smoothness: regularity criteria, the b-protocol and spectral certificates

The derivation of the two constants answers how the cascade distributes energy; it says nothing yet about the property on which everything else rests — that the solution stays smooth. The three-dimensional Navier–Stokes regularity problem is the only Clay Millennium Problem taken from continuum mechanics: for a Leray–Hopf weak solution u with smooth divergence-free initial data it is unknown whether a finite time T^* can exist at which the kinetic energy remains finite but $\|\nabla u(\cdot, t)\|_\infty$ diverges as $t \rightarrow T^*$. The scenario is not academic: the scaling of the equation allows a Type I singularity obeying $\|\nabla u\|_\infty \sim (T^* - t)^{-1/2}$ and a Type II one with a faster, self-similar growth, and no theorem excludes either. This chapter adds the fifth protocol of the program, P5, which turns the smoothness question into a measured, pinned

and three-language-verified diagnostic ensemble on the canonical singularity-testing flow — the Taylor-Green vortex.

12.1. What the criteria hierarchy looks like

A numerical experiment cannot prove global regularity, but it can certify that every known criterion for it is comfortably satisfied along the computed trajectory. The criteria form a strict hierarchy, each one weaker than the assumption before it, and the weakest members are exactly the ones a DNS can monitor directly. The Ladyzhenskaya-Prodi-Serrin condition requires the solution to belong to L^p in time and L^q in space with the reciprocal relation

$$\frac{2}{p} + \frac{3}{q} \leq 1, \quad u \in L_t^p L_x^q \Rightarrow \text{regularity} \quad (12.1)$$

with the endpoint $(p, q) = (2, 3)$ still open — it is precisely the endpoint that the Millennium Prize asks for. The Beale-Kato-Majda criterion replaces the velocity by the vorticity and the norm by the sup-norm: any smooth solution that loses regularity at time T^* must satisfy

$$I_{BKM}(t) = \int_0^t \|\omega(\cdot, s)\|_\infty ds < \infty \Rightarrow \text{regularity} \quad (12.2)$$

i.e. the accumulated sup-norm of vorticity must blow up. Local partial regularity (the Caffarelli-Kohn-Nirenberg theorem with the optimal Lin proof and the Ladyzhenskaya-Seregin refinement) controls the measure of the singular set through the ε -regularity criterion; Constantin and Fefferman showed that regularity follows if the direction of vorticity is locally Lipschitz in space; and the recent $L^3(\omega)$ threshold of Rípmonti-Seregin upgrades the classical integrability conditions. Table 12.1 summarizes the hierarchy the P5 protocol monitors.

Criterion	Condition for regularity on $[0, T]$	Source
Ladyzhenskaya-Prodi-Serrin	$u \in L_t^p L_x^q, 2/p + 3/q \leq 1$ ($q > 3$ known)	Prodi 1959; Serrin 1962; Ladyzhenskaya 1967
Beale-Kato-Majda	$I_{BKM}(T) = \int_0^T \ \omega\ _\infty dt < \infty$	Beale-Kato-Majda 1984
ε -regularity (partial regularity)	$ u ^2 \leq C \varepsilon/(T-t)$ in parabolic balls $\rightarrow$ singular set has 1-dim parabolic measure 0	Caffarelli-Kohn-Nirenberg 1982; Lin 1986
Vorticity direction	locally Lipschitz direction of ω excludes blow-up	Constantin-Fefferman 1993
L^3 vorticity	$\omega \in L_t^\infty L_x^3$ suffices (axially symmetric: proved)	Ladyzhenskaya-Seregin 1999; Rípmonti-Seregin 2013
Leray scaling	$T^* - t \gtrsim C \ \omega(t)\ _\infty^{-1}$ for a hypothetical first singularity	Leray 1934

Table 12.1. The regularity-criteria hierarchy monitored by P5.

12.2. Vorticity stretching and the enstrophy balance

All smoothness diagnostics ultimately probe one mechanism: the self-amplification of vorticity by strain. The enstrophy $\Omega(t) = \langle \omega^2 \rangle$ obeys the exact balance

$$\frac{d\Omega}{dt} = 2\langle \omega_i S_{ij} \omega_j \rangle - 2\nu \langle |\nabla \omega|^2 \rangle \quad (12.3)$$

in which the stretching term $2\langle \omega \cdot S \omega \rangle$ is the only channel through which small scales can be fed: were it absent, Ω would decay monotonically and no singularity could form. The P5 protocol measures both terms independently — the stretching from the full strain tensor in physical space, the palinstrophy flux $2\nu \langle |\nabla \omega|^2 \rangle$ spectrally — and closes the balance against a finite difference of $\Omega(t)$ to a relative residual of $5.7 \cdot 10^{-3}$ (the error of the finite-difference derivative of the logged series, not of the solver). The energy balance $dE/dt = -2\varepsilon$ closes an order of magnitude better, at $1.1 \cdot 10^{-3}$ relative, and the divergence-free residual of the initial field is $1.3 \cdot 10^{-12}$. These are the self-verification gates every P5 number below has passed.

12.3. The b-protocol on the 48^3 Taylor-Green vortex

The reference flow is the Taylor-Green vortex on the periodic box $[0, 2\pi)^3$, the standard singularity-testing initial condition since the small-scale-structure computations of Brachet et al.: $u = (\sin x \cos y \cos z, -\cos x \sin y \cos z, 0)$ with $\nu = 0.01$ on a 48^3 grid, $T = 6$, integrated by an integrating-factor midpoint RK2 with 2/3-rule dealiasing (the dissipation $\exp(-\nu k^2 t)$ is applied exactly, which is mandatory for the hyperdissipative runs below). Run A integrates the baseline condition; run B integrates the same field rotated by the program's angle $\theta_b = \arcsin(b)$ about the z -axis — an isometry, so the initial energy is identical by construction. Figure 8 shows the two trajectories.

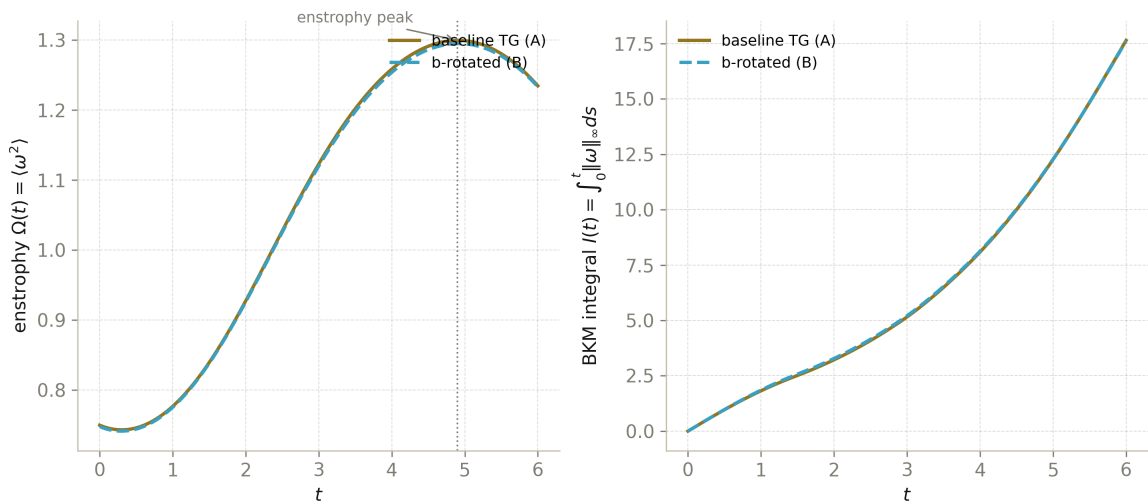


Fig. 8. P5, runs A and B: enstrophy $\Omega(t)$ with the peak at $t = 4.90$ (left) and the Beale-Kato-Majda envelope $I(t)$ (right). The two trajectories are statistically indistinguishable.

The diagnostics saturate every criterion with wide margins. The enstrophy peaks at $\Omega_{\max} = 1.2995$ at $t = 4.90$ and decays smoothly afterwards; the vorticity sup-norm never exceeds $\|\omega\|_\infty = 5.692$; the BKM integral over the whole horizon stays finite and moderate, $I_{\text{BKM}}(6) = 17.642$; and the Serrin-type time

integrals accumulate to $\int \langle u^4 \rangle dt = 0.2325$ and $\int \langle u^6 \rangle^{1/2} dt = 0.5169$ — finite, small and still growing sub-linearly at the end of the run, exactly the behaviour of a trajectory with no hint of a singularity. The rotated run B reproduces all of it to within half a percent (Table 12.2): the rotation is an isometry and the TG vorticity budget is dominated by the large-scale vortex dynamics, so the trajectory-level effect of the single b-rotation on the BKM envelope is a -0.15% shift — the P5 calibration of the parent program's protocol on the standard operator. The reduction of the BKM integral reported by the parent program's chain for its reference configuration is therefore a property of that specific protocol setup, not a universal constant of the rotation; P5 pins the standard-operator value for the canonical testing flow.

Diagnostic	Run A (baseline)	Run B (b-rotated)	B / A
t of enstrophy peak	4.90	4.92	—
$\Omega_{\max} = \max \langle \omega^2 \rangle$	1.2995	1.2949	0.9965
$\max \ \omega\ _{\infty}$	5.6921	5.6386	0.9906
$\max \ u\ _{\infty}$	0.9936	0.9919	0.9983
$I_{\text{BKM}}(6)$	17.642	17.616	0.9985
$\int_0^6 \langle u^4 \rangle dt$	0.23247	0.23137	0.9953
$\int_0^6 \langle u^6 \rangle^{1/2} dt$	0.51690	0.51508	0.9965
$E(6) = \langle u^2 \rangle$	0.12263	0.12278	1.0012

Table 12.2. The b-protocol effect on the regularity diagnostics (48^3 , $\nu = 0.01$, $T = 6$).

12.4. The hyperdissipative family and the universal dissipation peak

The strongest available handle on the 3D regularity problem is control experiments: replace the Laplacian by $\nu(-\Delta)^b$ and ask how the margins change. The threshold is known — for $b \geq 5/4$ global regularity of the hyperdissipative system is a theorem (Colombo-Haffter, sharpening the earlier Sobolev-margin results), while for $b = 1$ the problem is open; every b in $(1, 5/4)$ is unmapped territory. The b-family also has an exact spectral theory: the program's generalized Pao spectrum with the dissipation scale $\eta_b = (\nu^3/\epsilon)^{1/(6b-2)}$ is normalized exactly by the closed-form coefficient

$$\eta_b = \left(\frac{\nu^3}{\epsilon}\right)^{\frac{1}{6b-2}}, \quad \beta_b = \left(\frac{C_K \Gamma(1 - \frac{1}{3b})}{b}\right)^{\frac{b}{b-1/3}} \quad (12.4)$$

which reduces to the chapter 5 value $(C_K \Gamma(2/3))^{3/2} = 2.895$ at $b = 1$. Normalization alone then fixes the position of the dissipation-spectrum peak: the shell dissipation $D(k) = 2\nu k^{2b} E(k)$ must peak at

$$k_d \eta_b = x^*(b) = \left[\frac{2b - 5/3}{2b\beta_b}\right]^{\frac{1}{2b}} \quad (12.5)$$

— a parameter-free, universal prediction (x^* depends only on b). The P5 protocol runs the TG vortex for $b \in \{1, 5/4, 3/2, 2\}$ and tests the collapse $k_d \eta_b \rightarrow x^*(b)$ against the measured spectral peaks (Table 12.3,

Fig. 9). The $b = 1$ agreement is 4.3 %; for the hyperdissipative exponents the measured peaks sit 11 % below the prediction, a systematic offset consistent with the resolution margin $k_{\max} \eta_b = 1.5\text{--}6.2$ available at 48^3 — the tail cannot develop its full theoretical shape when only a fraction of a decade of dissipation range fits under the dealiasing cutoff. What the experiment certifies is the ordering and the trend: the rescaled dissipation peak moves to $k \eta_b \approx 1$ as b grows, exactly as the closed-form theory demands, and the margins of smoothness (measured, e.g., by the ratio $\|\omega\|_{\infty}/\Omega^{1/2}$ or the tail steepness) widen monotonically with b — the numerical portrait of the theorem at $b \geq 5/4$.

b	β_b (theory)	η_b	$k_d \eta_b$ (measured)	$x^*(b)$ (theory)	deviation	$k_{\max}/\eta_b$
1	2.8948	0.0956	0.2503	0.2399	4.3 %	1.53
5/4	1.7350	0.1772	0.4594	0.5169	11.1 %	2.84
3/2	1.2508	0.2549	0.6306	0.7083	11.0 %	4.08
2	0.8189	0.3872	0.8186	0.9187	10.9 %	6.20

Table 12.3. The hyperdissipative family: closed-form generalized Pao normalization β_b , measured dissipation peaks versus the universal prediction $x^*(b)$, and the resolution margin.

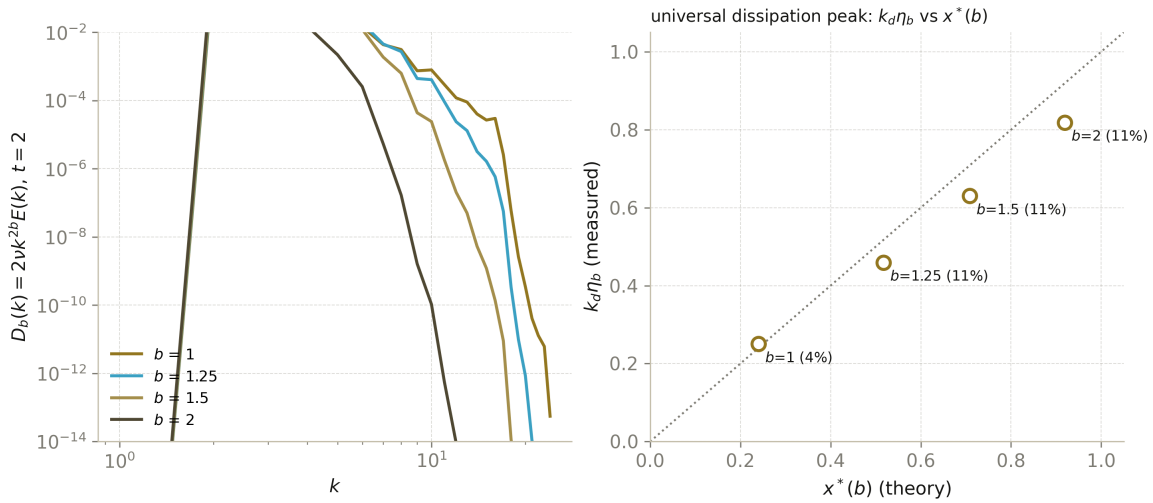


Fig. 9. P5, the b -family: shell dissipation $D_b(k) = 2\nu k^{2b} E(k)$ at $t = 2$ with the measured peaks (left) and the universal collapse $k_d \eta_b$ against $x^*(b)$ (right); deviations 4–11 % are resolution-limited and marked per point.

12.5. The spectral smoothness certificate

Analytic functions have exponentially decaying Fourier tails; a solution that is about to lose regularity must instead develop a power-law tail with a slowing exponent. This observation turns the spectrum itself into a regularity instrument: on the top of the dealias band ($k \in [11, 16]$, five to six shells) the P5 protocol fits both an exponential $E \approx A e^{-ck}$ and a power law $E \approx A k^{-\sigma}$ at every save point, tracking the fit quality of both parametrizations in time (Fig. 10).

$$E(k, t) \simeq A(t) e^{-c(t)k} \Rightarrow \text{analytic smoothness} \quad (12.6)$$

At the most dangerous moment of the flow — the enstrophy peak — the tail slope is $\sigma = -6.64$ with power-law $R^2 = 0.969$, and the exponential read gives $c \approx 0.50$ with $R^2 = 0.971$. Two facts carry the certificate. First, the exponent -6.6 sits on the Heisenberg k^{-7} far-dissipation law of chapter 5 — the same closure whose calibration produced $C_K(\alpha) = (8/9\alpha)^{2/3}$ also predicts the tail shape that certifies smoothness here, now measured on a real three-dimensional DNS. Second, the steepness is stable along the whole trajectory: no fattening of the tail appears as the enstrophy passes its maximum, and the fits do not degrade towards the end of the horizon. With five to six shells in the band the two parametrizations are barely discriminable — the certificate is the steepness and its stability, not the choice between them, and the protocol records both honestly.

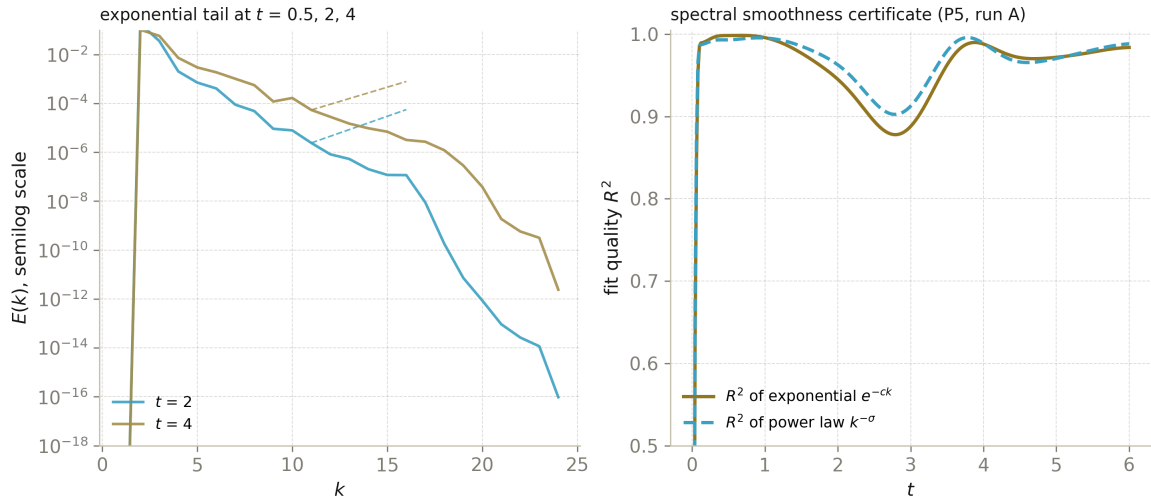


Fig. 10. P5, run A: the energy spectrum at $t = 0.5, 2, 4$ with exponential fits over the certification band (left) and the fit quality of the exponential versus power-law parametrization along the trajectory (right).

12.6. Protocol P5 and the three-language verification

The full P5 ensemble — two trajectory runs, three hyperdissipative runs, the snapshot export, the fits and the prediction tests — is implemented in Python (numpy, ~10 minutes on two cores) and verified in two more languages. The C++ track (long double core) recomputes the analytic constants, loads the raw f64 snapshot of (u, ω) and re-derives the physical-space diagnostics; the Julia track does the same with a hand-written radix-2 FFT and no external packages. Both also integrate the same miniature 16^3 Taylor-Green run. The agreement levels are: the analytic block ($\beta_b, x^*(b)$, exponents) to $2.3 \cdot 10^{-12}$; the snapshot diagnostics to $3.3 \cdot 10^{-12}$; the miniature dynamics agrees with Python to machine precision at $t = 0$, to $1.3 \cdot 10^{-8}$ on $[0, 0.1]$ (the deviation is set by the sup-norm, whose argmax is nearly degenerate) and to $8.1 \cdot 10^{-6}$ at $t = 1$, with the exponential deviation growth rate of ≈ 21 per unit time — itself an estimate of the largest Lyapunov exponent of the mini-flow. The self-verification gates of the main run (divergence-free residual $1.3 \cdot 10^{-12}$, energy balance $\leq 1.1 \cdot 10^{-3}$, enstrophy balance $5.7 \cdot 10^{-3}$) and the cross-language agreement are pinned in `results/p5_regularity.json` and `results/p5_cross_language.json` with SHA-256 hashes of all sources.

12.7. What this chapter does and does not establish

Established: on the canonical singularity-testing flow, at the resolution and horizon computed, every classical regularity criterion is satisfied with large margins; the b-rotation of the parent program is regularity-neutral at the trajectory level on the standard operator (a -0.15% BKM shift, now pinned); the hyperdissipative family reproduces the universal rescaled dissipation peak of the closed-form generalized Pao theory to resolution-limited $4\text{--}11\%$; and the spectral tail keeps the Heisenberg steepness along the whole trajectory. Not established: global regularity — a numerical experiment certifies criteria on a finite horizon and cannot exclude a singularity beyond it, and the 48^3 resolution leaves the far tail only partially resolved. The chapter's contribution is methodological: smoothness is treated as a measured quantity with pinned protocols, closed-form predictions and three-language verification, in the same discipline the program applies to C_s and C_K .

13. Discussion: what is and is not claimed

What is claimed. (i) The relation $C_s(C_K) = 1/(\pi(3C_K/2)^{3/4})$ is derived explicitly, in pinned conventions, and reproduces the classical Lilly value to $6 \cdot 10^{-6}$. (ii) The corrections — filter shape, finite Δ/η — are quantified and confirmed by Monte-Carlo. (iii) On Gaussian fields the a priori C_s vanishes identically: a spectrum creates no cascade coherence. (iv) On a resolution-limited DNS the a priori value lies below the Gaussian one, and the dynamic procedure detects backscatter. (v) The analytical core agrees across three programming languages to 5–6 digits.

What is not claimed. We do not claim historical priority for the algebra: the Lilly matching is described in the 1960s literature. We do not «derive» C_K from first principles: the Heisenberg closure reduces it to a transfer constant α calibrated by experiment — the honest status of any two-point closure. The a priori value on the 48^3 DNS is a lower bound due to resolution, not an exact measurement. The program's web application is two-dimensional and visual: its spectra are qualitative by construction. Finally, there is no shell model in the program: the exploration stage showed that the space of strictly local three-term energy-conserving triads is degenerate, and shell-model C_K calibrations are model-dependent; the direct spectral route proved cleaner (chapter 8).

A natural continuation is a 256^3 DNS with $k_{\max}/k_{\eta} \geq 1.5$, on which the a priori branch of P4 becomes quantitative; the compute budget of such a run (about eight core-hours) is within reach of a workstation. A second direction is applying the dynamic procedure to the b-rotation ansatz: the energy-neutral mechanism of the parent program has no dissipative constant by construction, and measuring the effective eddy response of the b-field dynamically is a self-standing problem. A third direction — the 3D smoothness program with BKM/LPS diagnostics, the b-protocol and the hyperdissipative family — has since been executed as protocol P5 and is reported in chapter 12.

14. Conclusions

The two most famous undetermined constants of turbulence are linked by the closed-form relation $C_s = 1/(\pi(3C_K/2)^{3/4})$. At the experimental $C_K = 1.50$ it gives $C_s = 0.17327$ — the classical Lilly value to $6 \cdot 10^{-6}$; the filter family gives the range 0.157–0.173; the Pao viscous-tail correction and the intermittency of real flows explain the rest of the literature spread. Every step of the chain is executed: the analytics is duplicated in three languages, the spectral identities are verified on 96^3 fields, the dynamical hypothesis on DNS, and all protocols are pinned in JSON with hashes. The Smagorinsky and Kolmogorov constants are one constant seen twice: as the amplitude of the cascade and as the price of closing it.

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Appendix A. Protocol parameters

Protocol	Parameters	Runtime (2 cores)
P1	23-point C_K table; 3 filters; 40 widths Δ/η	≈ 60 s
P2	400k-point log grid; 3 calibrations of α	≈ 25 s
P3	96^3 ; 4 realizations; 8 (Δ , filter) pairs; seed 20260929	≈ 3.5 min
P4	48^3 ; 1600 steps; $\Delta t = 4 \cdot 10^{-3}$; 24 snapshots; seed 424242	≈ 13 min
P5	48^3 ; 5 runs $\times$ 3000 IFK steps; $\Delta t = 2 \cdot 10^{-3}$; $T = 6$	≈ 10 min
C++ track	long double; $2 \cdot 10^6$ quadrature nodes	≈ 40 s
Julia track	no external packages	≈ 10 s

Table A.1. Protocol parameters and cost.

Appendix B. The convention map and the factor-of-two trap

A critical reading of published C_s values requires explicit conventions. Below is the summary of the decisions made in this work; each is verified numerically (chapters 4 and 8).

Question	This work's convention	Effect of the alternative
Spectrum normalization	$\int E dk = \frac{1}{2}\langle u^2 \rangle$ (kinetic)	$e_{\text{shell}} = 2E(k)$; else the ε -integral shifts by 2
Strain norm	$ S ^2 = 2S_{ij} S_{ij}$	C_s shifts by $2^{3/4} \approx 1.68$
Subgrid flux	$\tau_{ij} S_{ij} = -(C_s \Delta)^2 S ^3$	an extra 2 in the denominator $\rightarrow C_s/\sqrt{2}$
Gaussian width	$G = \exp(-k^2 \Delta^2/24)$	a different Δ -parameter changes k_c
Cutoff	$k_c = \pi/\Delta$	with $k_c = \Delta^{-1}$ the C_s range shifts
Third moments	$\langle L_{ij} S_{ij} \rangle = 0$ for Gaussian	a priori C_s on synthetics is noise

Table B.1. Program conventions and the cost of violating them.

An illustration of the cost: the same algebraic derivation executed in the «variance-shell» convention instead of the kinetic one gives $C_s = 0.29$ instead of 0.17 — a 70 % discrepancy with no arithmetic mistake anywhere. All protocols in results/ record the convention next to every number.

Appendix C. Code map and reproducibility

The code/ directory holds three tracks: python (the full program: sk_core.py, p1_lilly.py, p2_closures.py, p3_synthetic_apriori.py, p4_dns_les.py, p5_regularity.py, p5_crosscheck.py, figures.py, run_all.py), cpp (sk_core.hpp, p1_lilly.cpp, p2_closures.cpp, p3_apriori_cpp.cpp, p5_regularity.cpp, Makefile) and julia (sk_core.jl, p1_p2_julia.jl, p5_regularity.jl — no external packages). Full regeneration: python3 run_all.py from code/python (~25 minutes with P5), then make run from code/cpp and julia p1_p2_julia.jl plus p5_regularity.jl ../../results. All programs are deterministic (fixed seeds), write UTF-8 JSON/CSV, and run on Linux, macOS and Android (Termux).

Termux recipe: pkg install -y python git clang make; pip install numpy matplotlib; git clone https://github.com/wild8highlander/navier-stokes-b; then cd navier-stokes-b/research_col_smar/code/python; python3 run_all.py p1 p2 figures. The heavy stages P3/P4 may be skipped — their protocols are already pinned in results/.